WORKED SOLUTION

STEP III 2014, Question 5

Pure • Complex Numbers in Geometry

STEP III 2014, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q5

The key idea

In the complex plane, "parallel and equal in length" just means "equal as displacement vectors", and a quarter turn anticlockwise is multiplication by i. So every geometric condition in this question becomes a short linear equation in the vertices. For the squares on the sides, we find each centre by walking to the midpoint of the side and then stepping outwards by half the side length, at right angles to it.

Solution

The parallelogram criterion

Suppose first that $a + c = b + d$. Then

$$b - a = c - d \qquad \text{and} \qquad d - a = c - b,$$

so the displacement $\overrightarrow{AB}$ equals $\overrightarrow{DC}$ and $\overrightarrow{AD}$ equals $\overrightarrow{BC}$. Equal displacements are parallel and of equal length, so both pairs of opposite sides are parallel and equal: $ABCD$ is a parallelogram.

Conversely, suppose $ABCD$ is a parallelogram. Two segments that are parallel and of equal length have displacements that are equal or opposite, so

$$b - a = \varepsilon(c - d), \qquad d - a = \eta(c - b), \qquad \varepsilon, \eta \in \{1, -1\}.$$

If $\varepsilon = 1$ or $\eta = 1$, either equation rearranges to $a + c = b + d$. The only other case is $\varepsilon = \eta = -1$, that is $b - a = d - c$ and $d - a = b - c$. Subtracting the second from the first gives $b - d = d - b$, so $b = d$, which is impossible for distinct vertices. Hence $a + c = b + d$, and the equivalence is proved.

The square criterion

Let $ABCD$ be a parallelogram, and let $m = \frac{1}{2}(a+c) = \frac{1}{2}(b+d)$, the common midpoint of the diagonals (its centre). Then

$$a-m = \tfrac{1}{2}(a-c), \qquad b-m = \tfrac{1}{2}(b-d), \qquad c-m = -(a-m), \qquad d-m = -(b-m).$$

If $ABCD$ is a square, its diagonals are equal in length and bisect each other at right angles, so B is obtained from A by a quarter turn about m. That turn is anticlockwise because the vertices are labelled anticlockwise. So $b-m = \mathrm{i}(a-m)$, which is $\frac{1}{2}(b-d) = \frac{1}{2}\mathrm{i}(a-c)$, that is, $\mathrm{i}(a-c) = b-d$.

Conversely, if $\mathrm{i}(a-c) = b-d$, write $w = a-m \neq 0$. Then

$$a-m = w, \quad b-m = \mathrm{i}w, \quad c-m = -w, \quad d-m = -\mathrm{i}w,$$

so the four vertices are w turned successively through 90° about m. They are the vertices of a square (all four sides are $|w|\sqrt{2}$, and each side is the previous one multiplied by i). This proves the second equivalence.

Part (i)

The vertices P, Q, R, S run anticlockwise, so as we walk from P to Q the interior of $PQRS$ is on our left. The external square on PQ therefore lies on our right.

The centre X is reached from the midpoint $\frac{1}{2}(p+q)$ of PQ by moving a distance $\frac{1}{2}|q-p|$ (half the side) perpendicular to PQ, to the right. Multiplying $q-p$ by $-\mathrm{i}$ turns it through 90° clockwise, i.e. to the right, so this step is $-\frac{1}{2}\mathrm{i}(q-p)$. Hence

$$x = \tfrac{1}{2}(p+q) - \tfrac{1}{2}\mathrm{i}(q-p) = \tfrac{1}{2}(p(1+\mathrm{i}) + q(1-\mathrm{i})),$$

as required.

? **Ask yourself** Why $-\mathrm{i}$ rather than $+\mathrm{i}$?

Multiplying by $+\mathrm{i}$ would turn $\overrightarrow{PQ}$ to the left, which points into the quadrilateral and would give the centre of the square built on the inside. Because every internal angle is less than 180°, the interior lies entirely on the left of each side, so the right is genuinely outside.

Part (ii)

Applying part (i) to each side in turn (the same reasoning works for every side, since each is traversed with the interior on the left),

$$2x = p(1+\mathrm{i}) + q(1-\mathrm{i}), \qquad 2y = q(1+\mathrm{i}) + r(1-\mathrm{i}),$$
$$2z = r(1+\mathrm{i}) + s(1-\mathrm{i}), \qquad 2t = s(1+\mathrm{i}) + p(1-\mathrm{i}).$$

We compare the two expressions in the criteria.

Sums of opposite vertices.

$$\begin{aligned} 2(x+z)-2(y+t) &= (p+r)(1+\mathrm{i})+(q+s)(1-\mathrm{i})-(q+s)(1+\mathrm{i})-(p+r)(1-\mathrm{i}) \\ &= 2\mathrm{i}\,(p+r-q-s), \end{aligned}$$

so

$$x+z-(y+t) = \mathrm{i}(p+r-(q+s)). \tag{1}$$

Diagonals.

$$\begin{aligned} 2\mathrm{i}(x-z) &= \mathrm{i}(p-r)(1+\mathrm{i})+\mathrm{i}(q-s)(1-\mathrm{i}) = (p-r)(\mathrm{i}-1)+(q-s)(1+\mathrm{i}), \\ 2(y-t) &= (q-s)(1+\mathrm{i})+(r-p)(1-\mathrm{i}) = (q-s)(1+\mathrm{i})+(p-r)(\mathrm{i}-1). \end{aligned}$$

These are identical, so

$$\mathrm{i}(x-z) = y-t \tag{2}$$

for *every* such quadrilateral $PQRS$: the segments XZ and YT are always equal in length and perpendicular.

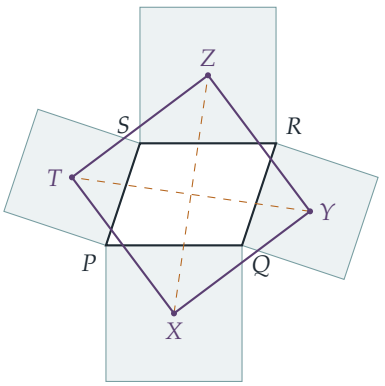
? **Ask yourself** Since (2) always holds, what decides whether $XYZT$ is a square?

Only the parallelogram condition. Now we put the pieces together.

If $PQRS$ is a parallelogram, then $p+r = q+s$, so by (1) $x+z = y+t$ and $XYZT$ is a parallelogram by the first result. Together with (2), the square criterion shows that $XYZT$ is a square. (The converse half of that criterion does not need any assumption about the labelling, and $X \neq Z$, since $x = z$ would mean $r-p = \mathrm{i}(s-q)$, so that $\overrightarrow{PR}$ would be $\overrightarrow{QS}$ turned a quarter turn *anticlockwise*; but in a convex quadrilateral labelled anticlockwise, $\overrightarrow{PR}$ is always $\overrightarrow{QS}$ turned *clockwise* through an angle between $0°$ and $180°$ and rescaled.)

If $XYZT$ is a square, then in particular it is a parallelogram, so $x+z = y+t$ by the first result. By (1), $\mathrm{i}(p+r-q-s) = 0$, so $p+r = q+s$ and $PQRS$ is a parallelogram.

Hence $XYZT$ is a square if and only if $PQRS$ is a parallelogram.



Checks

Take the parallelogram $p = 0$, $q = 4$, $r = 5 + 3\mathrm{i}$, $s = 1 + 3\mathrm{i}$ (drawn above). The formula gives $x = 2 - 2\mathrm{i}$, $y = 6 + \mathrm{i}$, $z = 3 + 5\mathrm{i}$, $t = -1 + 2\mathrm{i}$. The sides of $XYZT$ are $4 + 3\mathrm{i}$, $-3 + 4\mathrm{i}$, $-4 - 3\mathrm{i}$, $3 - 4\mathrm{i}$, each i times the previous one: a square of side 5. As a check on part (i), $x = 2 - 2\mathrm{i}$ is indeed the centre of the square with vertices 0, 4, $4 - 4\mathrm{i}$, $-4\mathrm{i}$ below PQ. A computer check on thousands of random convex quadrilaterals confirmed identity (2) every time, and that $XYZT$ was a square exactly when $PQRS$ was a parallelogram.

Common slips

- Using $+\mathrm{i}$ in part (i), which puts the squares inside the quadrilateral.
- In the parallelogram criterion, forgetting that "parallel and equal" allows $b - a = -(c - d)$, which has to be ruled out.
- Proving only one direction of an "if and only if". In part (ii) the direction "square $\Rightarrow$ parallelogram" needs only (1).

Beyond STEP

Identity (2) is van Aubel's theorem: for *any* quadrilateral, the segments joining the centres of opposite external squares are equal and perpendicular. It is a small example of a general principle: geometric constructions built from rotations and midpoints are linear maps on the vertices, so theorems about them reduce to algebraic identities (Napoleon's theorem on equilateral triangles is another). Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully.

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